

Analysis of Variance (ANOVA):

ANOVA is a statistical technique used to compare the means of two or more populations by breaking the total variation of data set into different variance components.

ANOVA is used for two different purposes

- i) to estimate and test hypothesis about population means
- ii) to estimate and test hypothesis about population variances.

Complete Randomized Design (CRD) / (One Way Classification):

A design of experiment in which each unit can be allocated to one of the several groups is called CRD. In other words, a design where the data consists of independent random samples from each of K populations is called CRD.

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Data structure:

Samples $\longrightarrow$

	Treatment 1	Treatment 2	...	Treatment i	...	Treatment K	
	X_{11}	X_{21}	...	X_{i1}	...	X_{K1}	
Observations	X_{12}	X_{22}	...	X_{i2}	...	X_{K2}	
	$\vdots$	$\vdots$		$\vdots$		$\vdots$	
	X_{1n_1}	X_{2n_2}	...	X_{in_i}	...	X_{Kn_K}	
Total	$T_{1.}$	$T_{2.}$	...	$T_{i.}$	...	$T_{K.}$	$T_{..}$
	$\bar{X}_1$	$\bar{X}_2$	...	$\bar{X}_i$	...	$\bar{X}_K$	$\bar{X}_{..}$

Response of
Experimental Units

Each observation may be written in the form

$$X_{ij} = \mu_i + \epsilon_{ij} \quad ; \quad i = 1, 2, \dots, K$$

$$j = 1, 2, \dots, n_i$$

$$\text{error} \leftarrow \epsilon_{ij} = X_{ij} - \mu_i$$

We define the Population mean $\mu = \frac{\sum_{i=1}^K \mu_i}{K}$

$$\mu_i = \mu + \tau_i$$

then

$$X_{ij} = \mu + \tau_i + \epsilon_{ij} \quad ; \quad \sum_{i=1}^K \tau_i = \sum_{i=1}^K (\mu_i - \mu) = 0$$

Assumptions:

- i) The data constitute K independent random samples population
- ii) Each sample comes from normal with possibly different

~~variance~~ mean but constant variance under the model: $X_{ij} = \mu + \tau_i + \epsilon_{ij}$; $i=1, 2, \dots, K$; $j=1, 2, \dots, n_i$

Samples

- iii) X_{ij} are uncorrelated in pairs.

Hypotheses:

Null hypothesis: $H_0: \mu_1 = \mu_2 = \dots = \mu_K$

Alternative hypothesis: H_A : At least two means are unequal

Total Variation = Variation Between Group + Variation Within Group

$$SST = SSB + SSW$$

$$\sum_{i=1}^K \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{..})^2 = \sum_{i=1}^K n_i (\bar{X}_i - \bar{X}_{..})^2 + \sum_{i=1}^K \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

The break up of degrees of freedom's

$$N-1 = K-1 + N-K$$

on simplification

$$SST = X_{11}^2 + X_{12}^2 + \dots + X_{K n_K}^2 - \frac{C^2}{N} \quad \text{correction term}$$

$$SSB = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \dots + \frac{T_K^2}{n_K} - C.T \quad C.T = \frac{T_{..}^2}{N}$$

$N = n_1 + n_2 + \dots + n_K$

$$SSW = SST - SSB$$

ANOVA Table

Source of Variation	Sum of Squares	Degrees of freedom	Mean Squares
Between treatment	SSB	$K-1$	$M_b = \frac{SSB}{K-1}$
Within treatment	SSW	$N-K$	$M_r = \frac{SSW}{N-K}$
Total	SST	$N-1$	

$$\frac{M_b}{M_r} = F$$

- a) If our assumptions about the observations are correct then we would expect $M_b > M_r$.
If $M_b > M_r$, then the statistics used for testing variances is F statistics

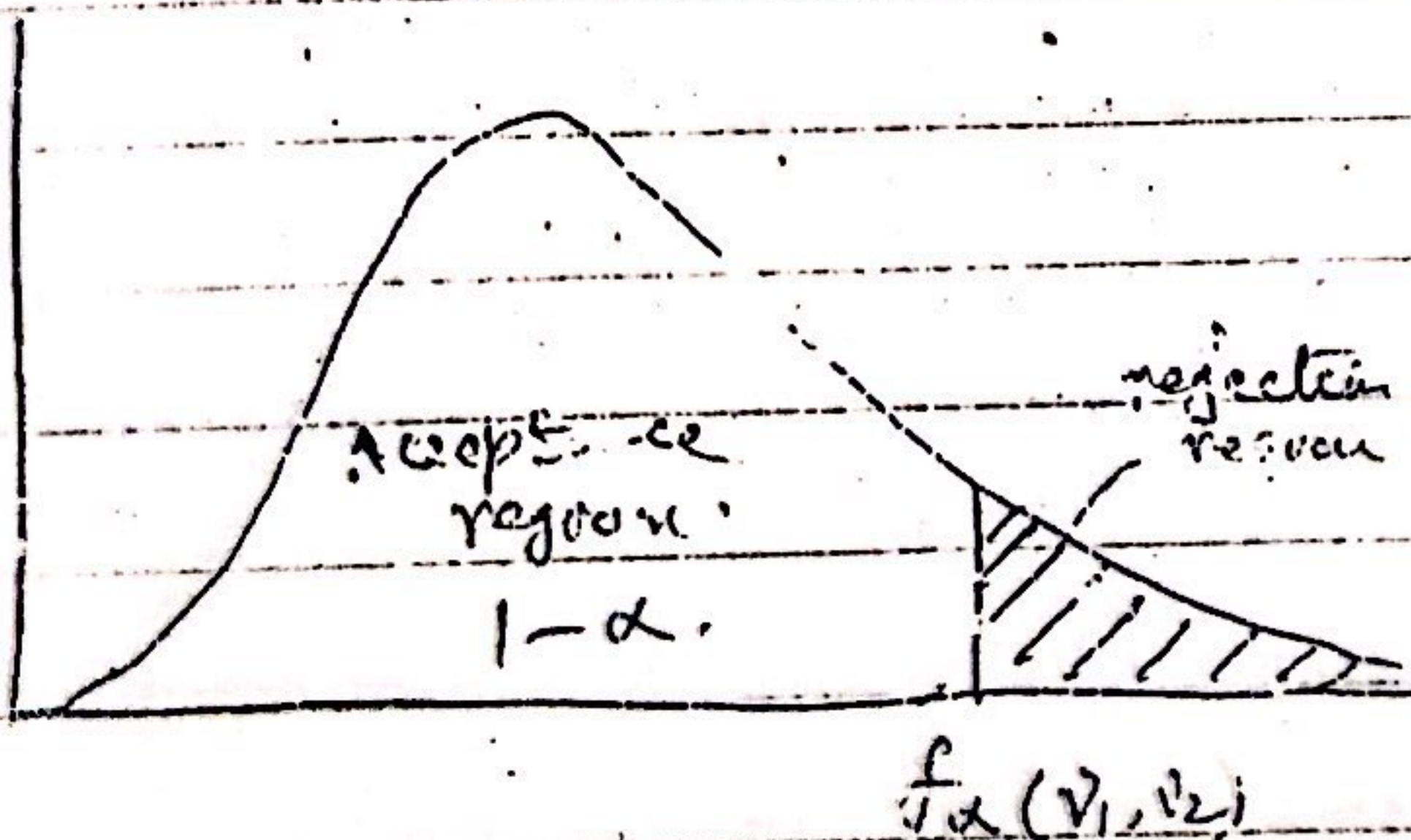
$$F_{\text{calculated}} = \frac{M_b}{M_r}; \quad F \sim f_{\alpha}(v_1, v_2)$$

F-distribution

If $F_{\text{calculated}} < F_{\alpha}(v_1, v_2)$, then we accept our null hypothesis and conclude that K treatments are equally effective.

- b) In the event of $M_b < M_r$, the data requires re-examination.

- c) If $M_b > M_r$ but $M_r < 1$ then f statistic may not be used no conclusion can be drawn.



Solution # 1:

1- Assumptions:

2- Model

$$X_{ij} = \mu + \tau_i + \epsilon_{ij}$$

$\swarrow$ $\downarrow$ $\searrow$ $\rightarrow$
 jth observation on i-th treatment overall mean i-th row effect error

$i = 1, 2, \dots, k$
 $j = 1, 2, \dots, n_i$

3- Hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_A : At least two of them are not equal.

4- Level of significance: $\alpha = 0.05$

5- Test statistics

$$F = \frac{S_1^2}{S_2^2} \sim F_{\alpha}(v_1, v_2)$$

6- Computations:

	Concentration				
	1	2	3	4	
	8.2	7.7	6.9	6.8	
	8.7	8.4	5.8	7.3	
	9.4	8.6	7.2	6.3	
	9.2	8.1	6.8	6.9	
		8.0	7.4	7.1	
			6.1		
Total	35.5	40.8	40.2	31.4	150.9
mean	8.875	8.16	6.7	6.88	7.545

$$C.T = \frac{(T_{..})^2}{N} = \frac{(150.9)^2}{20} = \frac{22770.81}{20} = 1138.54$$

$N = n_1 + n_2 + n_3 + n_4$

$$N = \sum n_i$$

$$SST = \sum_{i=1}^k \sum_{j=1}^n x_{ij}^2 - C.T$$

$$= (8.2)^2 + (7.7)^2 + (6.9)^2 + (6.8)^2 + \dots + (6.1)^2 - 1138.54$$

$$= 1157.89 - 1138.54$$

$$SST = 19.3495$$

$$SS \text{ Treatment} = \sum_{i=1}^k \frac{T_i^2}{n_i} - C.T$$

$$= \frac{(35.5)^2}{4} + \frac{(40.8)^2}{5} + \frac{(40.2)^2}{6} + \frac{(34.4)^2}{5} - 1138.54$$

$$= 1154.0025 - 1138.54$$

$$SS \text{ Treatment} = 15.4620$$

$$SS E = SST - SS \text{ Treatment}$$

$$= 19.3495 - 15.4620$$

$$SS E = 3.8875$$

ANOVA TABLE:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F Calculated
Between Treatment	15.4620	3	$5.1540 = \frac{15.4620}{3}$	
Within Treatment	3.8875	16	$0.2430 = \frac{3.8875}{16}$	
Total	19.3495	19		

$\frac{5.1540}{0.2430} = 21.2098$
 ≈ 21.21

Randomized Complete Block Design (RBD) [Two Way CLASSIFICATION]

The randomized block design is a design in which the units (called experimental units) to which the treatments are applied are subdivided into homogeneous groups called blocks, so that the number of experimental units in a block is equal to the number of treatments being studied. The treatments are assigned at random to the experimental units within each block, and every block receives every treatment. A typical form is

Data Structure:

Blocks	Treatments $\longrightarrow$					Total	Mean	
	1	2	...	j	...			k
1	X_{11}	X_{12}	...	X_{1j}	...	X_{1k}	$T_{1.}$	$\bar{X}_{1.}$
2	X_{21}	X_{22}	...	X_{2j}	...	X_{2k}	$T_{2.}$	$\bar{X}_{2.}$
...	...	...	...	...	...	...	...	...
i	X_{i1}	X_{i2}	...	X_{ij}	...	X_{ik}	$T_{i.}$	$\bar{X}_{i.}$
...	...	...	...	...	...	...	...	...
n	X_{n1}	X_{n2}	...	X_{nj}	...	X_{nk}	$T_{n.}$	$\bar{X}_{n.}$
Total	$T_{.1}$	$T_{.2}$	...	$T_{.j}$	...	$T_{.k}$	$T_{..}$	
Mean	$\bar{X}_{.1}$	$\bar{X}_{.2}$	...	$\bar{X}_{.j}$	...	$\bar{X}_{.k}$		$\bar{X}_{..}$

Each observations may be written in the form

$$X_{ij} = \mu_{ij} + \epsilon_{ij} \quad i = 1, 2, \dots, n$$

$$j = 1, 2, \dots, k$$

and

$$X_{ij} = \mu + \beta_i + \tau_j + \epsilon_{ij} \quad \text{error}$$

$\downarrow$ Block effect or row effect $\downarrow$ Treatment effect or column effect

The Population mean $= \mu = \frac{\sum_{i=1}^n \sum_{j=1}^k \mu_{ij}}{nk}$

$\mu_{i.} = \frac{\sum_{j=1}^k \mu_{ij}}{k}$ and $\mu_{.j} = \frac{\sum_{i=1}^n \mu_{ij}}{n}$

Assumptions: The observations X_{ij}

- 1) Comes from normal populations (independent)
- 2) Comes from populations with same variance
- 3) have possibly different means.

Hypotheses:

1. H_0' : $\mu_{1.} = \mu_{2.} = \dots = \mu_{n.}$

H_A' : At least two of them are not equal

2. H_0'' : $\mu_{.1} = \mu_{.2} = \dots = \mu_{.k}$

H_A'' : At least two of them are not equal

$SST = SS(\text{Blocks}) + SS(\text{Treatments}) + SS(\text{Error or Residual})$

$$\sum_{i=1}^n \sum_{j=1}^k (X_{ij} - \bar{X}_{..})^2 = k \sum_{i=1}^n (\bar{X}_{i.} - \bar{X}_{..})^2 + n \sum_{j=1}^k (\bar{X}_{.j} - \bar{X}_{..})^2 + \sum_{i=1}^n \sum_{j=1}^k (X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{..})^2$$

$SST = \sum_{i=1}^n \sum_{j=1}^k X_{ij}^2 - C.T$; $C.T = \frac{T_{..}^2}{nk}$

Row
 $SS(\text{Block}) = \sum_{i=1}^n \frac{T_{i.}^2}{k} - C.T$

Column
 $SS(\text{Treatment}) = \sum_{j=1}^k \frac{T_{.j}^2}{n} - C.T$

$SS \text{ Error or Residual} = SST - SS \text{ Blocks} - SS \text{ Treatment}$

D.F.

$nk - 1 = (n-1) + (k-1) + (n-1)(k-1)$

Total

Blocks

Treatments

Error or Residuals

$N = nk$

415-0
= 90.0

P-4

Solution # 2:

1 - Assumptions:

2 - Model:

$$X_{ij} = \mu + \beta_i + \tau_j + e_{ij}$$

X_{ij} : i th block observation when j th treatment is applied.
 μ : overall mean
 β_i : i th block effect
 τ_j : j th treatment effect
 e_{ij} : error
 $i = 1, 2, \dots, M$
 $j = 1, 2, \dots, K$

3 - Hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

H_A : At least two means are not equal

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_A : At least two means are not equal

4 - Level of significance: $\alpha = 0.05$

5 - Test statistics: (Variance Ratio Statistics)

$$F = \frac{S_1^2}{S_3^2} \quad F \sim f_{\alpha}(v_1, v_2)$$

6 - Computations:

Subject	Treatments				Total	Mean
	1	2	3	4		
1	12.8	11.7	11.5	12.6	48.6	12.15
2	10.6	14.2	14.7	16.5	56.0	14.00
3	11.7	11.8	13.6	15.4	52.5	13.13
4	10.7	9.9	10.7	9.6	40.9	10.23
5	11.0	13.8	15.9	17.1	57.8	14.45
Total	56.8	61.4	66.4	71.2	255.8	
Mean	11.36	12.28	13.28	14.24		12.79

ANOVA TABLE:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F Computed
Between subjects	45.28	4	11.32	4.86
Between treatments	23.24	3	7.75	3.34
Error	27.94	12	2.33	
Total	96.46	19		

7- Decision:

a) Reject H_0 if

$$F_{cal} > F_{0.05, (4, 12)} = 5.91 \quad 3.26$$

b) Reject H_0 if

$$F_{cal} > F_{0.05, (3, 12)} = 8.74 \quad 3.49$$

8- Conclusion:

a) Our computed value of f (between subjects)is less than the table value of $F_{0.05} (4, 12) = 5.91$,so we accept our null hypothesis and conclude that all ^{population} means are equal.b) Our computed value of f (between treatments)is less than the table value of $F_{0.05} (3, 12) = 8.74$,so we accept H_0 and conclude that all population means are equal.

$$C.T. = \frac{T_{..}^2}{nk} = \frac{(255.8)^2}{5 \times 4} = \frac{65433.64}{20} = 3271.682$$

$$SST = \sum_{i=1}^n \sum_{j=1}^k X_{ij}^2 - C.T.$$

$$= (12.8)^2 + (11.7)^2 + (11.5)^2 + (12.6)^2 + (10.6)^2 + \dots + (17.1)^2 - 3271.682$$

$$= 3368.14 - 3271.68$$

$$SST = 96.46$$

$$SS(\text{Subject}) = \sum_{i=1}^n \frac{T_{i.}^2}{k} - C.T.$$

$$= \frac{(48.6)^2 + (56)^2 + (52.5)^2 + (40.9)^2 + (57.8)^2}{4} - 3271.68$$

$$= \frac{13267.68}{4} - 3271.68$$

$$= \cancel{3316.92} - 3271.68$$

$$= 3316.92 - 3271.68$$

$$SS(\text{Subject}) = 45.24$$

$$SS(\text{Treatment}) = \sum_{j=1}^k \frac{T_{.j}^2}{n} - C.T.$$

$$= \frac{(56.8)^2 + (61.4)^2 + (66.4)^2 + (71.2)^2}{5} - 3271.68$$

$$= \frac{16474.6}{5} - 3271.68$$

$$= 3294.92 - 3271.68$$

$$SS(\text{Treatment}) = 23.24$$

$$SS \text{ Error} = SST - SS(\text{Subject}) - SS(\text{Treatment})$$

$$= 96.46 - 45.24 - 23.24$$

$$SS \text{ Error} = 27.98$$